Practice Problems:
Units, Voltage and Current, and Basic Circuit Elements

Introduction to Electric Circuits
Supplemental Instruction

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January 20 2026

1 Engineering Notation

Convert the following quantities to engineering notation (remember that the coefficient should be between 1 and 999):

(a) 1947 W

(b) 0.0025 A

(c) 0.1738 mJ

(d) 47946 nV

(e) 0.00092 GΩ

Solution

(a) 1.947 kW

(b) 2.5 mA

(c) 173.8 μJ

(d) 47.946 μV

(e) 920 kΩ

2 Electron Charge

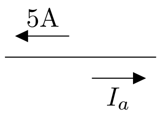
Sam's Superb Subatomic Particle Supply™ sells electrons for 5 n$ (nanodollars) each. How much would one Coulomb's worth cost at this rate?

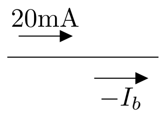
Solution

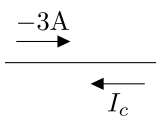
$1\text{C} \cdot \frac{\text{electron}}{1.6*10^{-19}\text{C}} \cdot \frac{\$5\cdot 10^{-9}}{\text{electron}} = 3.125 * 10^{10} = 31.25\ G\$$ (Gigadollars)

3 Current Notation

For each case, find the indicated current.

(a) 

(b) 

(c) 

Solution

(a) $I_a = -5\text{A}$

(b) $I_b = -20\text{mA}$

(c) $I_c = 3\text{A}$

4 Finding Current From Charge

If the charge passing from point at **A** to point **B** at some time t is described by $Q_{AB} = 3t^2 - e^{-2t}$, find the current from **B** to **A** at time $t = 0.5$s.

Solution

$$I_{BA} = -I_{AB} = -\left[\frac{dQ_{AB}}{dt}\right] = -\left[\frac{d}{dt}(3t^2 - e^{-2t})\right] = -\left[6t + 2e^{-2t}\right] = -6t - 2e^{-2t}$$

At time $t = 0.5$s,

$$I_{BA}(5) = -6(0.5) - 2e^{-2(0.5)} = -3.74\text{A}$$

5 Finding Charge From Current

When Noah Barlow's gaming rig boots up, the current it draws is described by

$$I = \sqrt{t+2}$$

Find the total charge that flows to the gaming rig from $t = 0$s to $t = 10$s.

Solution

$$Q = \int I dt = \int_0^{10} \sqrt{t+2} = \left[\frac{2}{3}(x+2)^{\frac{3}{2}}\right]_0^{10} = \frac{2}{3}(10+2)^{\frac{3}{2}} - \frac{2}{3}(0+2)^{\frac{3}{2}} = 25.83 \text{ C}$$

6 Voltage

A certain amount of work is done to move one Coulomb of charge from point **C** to point **D**. Afterwards, the voltage between **C** and **D** is 5V. How much work was done?

Solution

$$W = QV = 1\text{ C} \cdot 5\text{ V} = 5J$$

7 Voltage Notation

For each case, find the indicated voltage. Both voltages are between the same pair of points.

(a)
$$\begin{array}{ccccc} & - & V_x & + & \\ \overset{a}{\bullet} & + & 5\text{ V} & - & \overset{b}{\bullet} \end{array}$$

(b)
$$\begin{array}{ccccc} & + & V_y & - & \\ \overset{c}{\bullet} & - & -10\text{ V} & + & \overset{d}{\bullet} \end{array}$$

(c)
$$\begin{array}{ccccc} \overset{e}{\bullet} & + & 3\text{ mV} & - & \overset{f}{\bullet} \end{array}$$ Find V_{fe}.

Solution

(a) $V_x = -5$ V

(b) $V_y = 10$ V

(c) $V_{fe} = -3$ mV

8 Resistance and Conductance

Find the conductance of a resistor of value 1.5kΩ.

Solution

$$G = \frac{1}{R} = \frac{1}{1.5\text{k}\Omega} = 0.000667\text{ S} = 667\ \mu\text{S}$$

9 Resistor Color Codes

In each case, find the value and tolerance of the resistor with the specified color codes. Note that the codes may be read either from right to left or from left to right. Use the reading direction that gives a valid resistor value.

(a) Brown, Black, Red, Gold

(b) Silver, Orange, Purple, Yellow

(c) Orange, Orange, Brown, Green

Solution

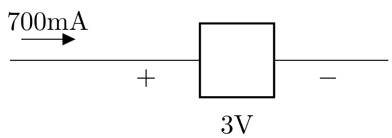
(a) Reading left to right: $('1' +' 0') \cdot 100\Omega = 10 \cdot 100\Omega = 1\text{k}\Omega \pm 5\%$

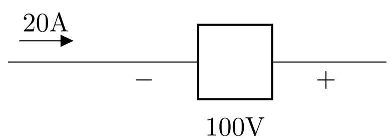
(b) Reading right to left: $('4' +' 7') \cdot 1\text{k}\Omega = 47 \cdot 1\text{k}\Omega = 47\text{k}\Omega \pm 10\%$

(c) Reading left to right: $('3' +' 3') \cdot 10\Omega = 33 \cdot 10\Omega = 330\Omega \pm 0.5\%$

10 Power

Find the absorbed power.

(a) 

(b) 

Solution

(a) $P = IV = 700\text{mA} \cdot 3\text{V} = 2.1\ \text{W}$

(b) $P = -IV = -20\text{A} \cdot 100\text{V} = -2\ \text{kW}$

11 Power and Energy

If a space heater uses 1500 W, how long will it take to transfer 750 kJ of energy to the room it's heating (assuming 100% efficiency)?

Solution

$P = \frac{E}{t} \Rightarrow t = \frac{E}{P} = \frac{750\text{kJ}}{1500W} = 500\ \text{s} = 8.3\ \text{minutes}$

12 Power

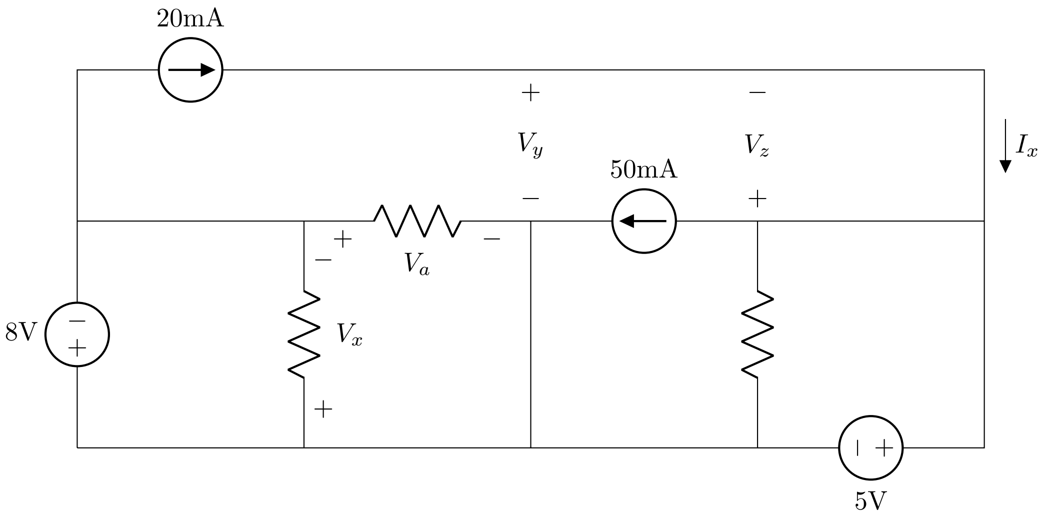
A resistor carries 2A of current and dissapates 600 J of energy in one minute. Find (a) the minimum required power rating of the resistor and (b) its resistance.

Solution

(a) $P = \frac{E}{t} = \frac{600J}{60s} = 10W$
(b) $P = IV = I(IR) = I^2R \Rightarrow R = \frac{P}{I^2} = \frac{10\text{W}}{(2\text{A})^2} = \frac{10\text{V}\cdot\text{A}}{4\text{A}^2} = 2.5\frac{\text{V}}{\text{A}} = 2.5\ \Omega$

13 Circuit Elements

Find V_x, V_y, V_z, V_a, and I_x.



Solution

$V_x = 8$ V
$V_y = 5$ V
$V_z = 0$ V
$V_a = -8$ V
$I_x = 20$ mA

14 Series and Parallel Components

For each pair of components, indicate whether the components are in series, parallel, or neither.

(a) R_7 and R_2

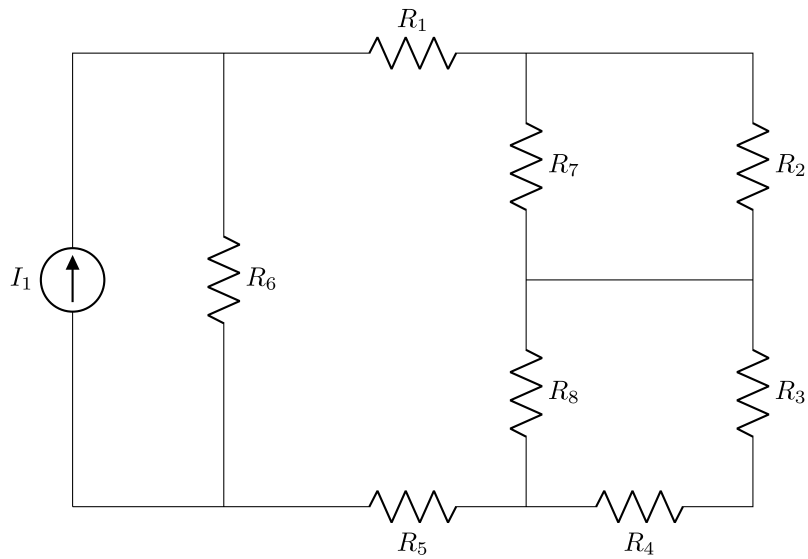
(b) I_1 and R_1

(c) R_8 and R_3

(d) R_3 and R_4

(e) R_1 and R_5

(f) I_1 and R_6



Solution

(a) R_7 and R_2 are in Parallel

(b) I_1 and R_1 are Neither

(c) R_8 and R_3 are Neither

(d) R_3 and R_4 are in Series

(e) R_1 and R_5 are in Series

(f) I_1 and R_6 are in Parallel

References

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- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
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